PORTFOLIO REPORT · R MARKDOWN

U.S. Criteria Air Pollutants, 2021–2025

An EPA AQS Statistical Analysis in R

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DATE
13 November 2026

DATA
EPA AQS

CODE
GitHub repository

2021–2025	5	16	3
STUDY YEARS	POLLUTANTS	HOURLY MONITORS	SITE GROUPS

CONTENTS

01 Introduction . . . 2
02 Data and methods . . . 3
03 National context . . . 4
04 Houston case study: daily . . . 6
05 Houston case study: hourly . . . 7
06 Distributions . . . 9
07 Limitations . . . 11
08 Conclusion and regulatory implications . . . 12
References . . . 13
Appendix . . . 13

REPRODUCIBILITY

Data: U.S. EPA Air Quality System, file versions dated 25 June 2026. Analysis done in R 4.6.1 and R Markdown, with package versions saved in renv.lock. The numbers in the text are pulled straight from output tables written by the analysis scripts, and all figures are made in code, so the full report can be rebuilt from the raw data. Code: github.com/Yousef-sham5/houston-dfw-air-quality.

01 Introduction

The U.S. Environmental Protection Agency (EPA) sets National Ambient Air Quality Standards (NAAQS) for six criteria pollutants, with limits chosen to protect public health [1]. Many areas still do not meet them. In 2023, 42.7% of ozone monitoring sites and 39.0% of annual PM2.5 sites with data in all five study years were above their standards (Section 3).

Houston and Dallas–Fort Worth both have long-running ozone problems. EPA classifies both metros as severe nonattainment areas for the 2008 ozone standard and serious nonattainment areas for the 2015 standard [2, 3]. In this dataset they rank 6th and 8th out of 353 metros with an ozone exceedance, ranked by exceedance site-days. That count grows with the number of monitors a metro has, so it is only a rough comparison. The biggest difference between the two metros is the Houston Ship Channel. The channel and the area around it support what the American Society of Civil Engineers calls the second largest petrochemical complex in the world [4], with refineries, chemical plants and shipping concentrated along its banks.

The goal of this study is to measure how air near the Ship Channel compares with the rest of Harris County ("Houston other") and with Dallas–Fort Worth, and to check whether the differences match what the chemistry predicts. Three hypotheses are tested:

> **H1.** NO_2 is higher near the channel. Combustion sources such as process heaters, flares, ship engines and diesel trucks all emit nitrogen oxides, so sites close to a dense cluster of them should read higher.
>
> **H2.** Ozone is lower near the channel because fresh NO reacts with it: $NO + O_3 \rightarrow NO_2 + O_2$. Each ozone molecule used up makes one NO_2 molecule, so the sum $Ox = O_3 + NO_2$ is conserved by this reaction. If titration explains the ozone gap, Ox should be about the same in both groups of sites.
>
> **H3.** SO_2 comes from short, local plumes. If so, it should show up as brief spikes that line up with certain wind directions, while the background stays near the detection limit.

The analysis uses EPA Air Quality System (AQS) data from 2021 to 2025 at two scales. Daily data for PM2.5, ozone, NO_2, SO_2 and CO across the U.S. give the national picture. For the case study, hourly ozone, NO_2 and SO_2 from 16 monitors in Houston and Dallas–Fort Worth are paired with hourly wind and temperature. The Ship Channel group is defined as the 9 Harris County sites within 7.5 km of the channel's land-cut sections. The hourly groups are small, with two ozone and NO_2 sites each, so the hourly results describe those specific sites. Each comparison is repeated with different site groupings and data subsets to check that it holds up. Because this is observational data, links to a source are stated as "consistent with" that source.

Section 2 covers the data and methods. Section 3 gives the national context, and Sections 4 and 5 present the daily and hourly case study. Section 6 describes how daily values are distributed, which curves fit them and how well those curves count high days, and works one ozone design value through EPA's rules to check it against EPA's published value. Section 7 lists the limitations, and Section 8 comes back to H1 to H3.

02 Data and methods

02.1 Data and cleaning

Data come from EPA's AirData pre-generated files [5], in the versions dated 25 June 2026: 25 daily, 15 hourly pollutant and 10 hourly weather files. Table 2.1 lists the parameter codes, daily values and standards. PM2.5 is limited to federal reference and equivalent methods (FRM and FEM), and the weather data are AQS wind speed, wind direction and temperature (61103, 61104, 62101).

Table 2.1. Pollutants, AQS parameter codes, daily values and the NAAQS each site-year was screened against [1].

Pollutant	AQS code	Daily value	Standard screened
PM2.5	88101	24-hour mean	35 µg/m^3, 98th percentile day
PM2.5 (annual)	88101	Annual mean	9.0 µg/m^3, mean of four quarters
Ozone	44201	Maximum 8-hour mean	0.070 ppm, 4th-highest day
NO_2	42602	Maximum 1-hour value	100 ppb, 98th percentile day
SO_2	42401	Maximum 1-hour value	75 ppb, 99th percentile day
CO	42101	Maximum 8-hour mean	9 ppm, 2nd-highest day

Cleaning took the daily files from 8,930,992 to 5,557,574 rows, mostly by removing duplicate summaries of the same monitor-day (Table A.1, Appendix). Gaps are never filled in, and daily values from days with under 75% of hours are dropped. Other unusual data, such as event days, hourly readings below the detection limit or on incomplete days, one monitor relocation and several method changes, are flagged and kept so each result can be rerun without them. Events are labelled from a sourced table of 203 state and county rows. The 79,402,201 hourly rows were read in chunks down to 661,545 monitor-hours for the 16 case-study monitors, and those hours rebuild all 27,747 daily maxima in EPA's daily files exactly. Times stay in local standard time, as AQS reports them.

02.2 Site groups

The Ship Channel group is the Harris County sites within 7.5 km of the channel's 12 land-cut sections in the U.S. Army Corps of Engineers channel framework [6]. The cutoff falls between Houston Deer Park #2 (6.94 km) and Houston North Wayside (8.49 km). The rest of Harris County is "Houston other", except Seabrook, which sits next to the separate Bayport industrial district. Dallas and Tarrant counties form Dallas–Fort Worth. Table 2.2 gives the sites in each group, and Section 4 maps the hourly sites.

Table 2.2. Sites per group. Daily counts are sites with data in all five years; hourly counts are the case-study monitors, which do not include PM2.5.

	Daily, five-year sites			Hourly monitors		
Pollutant	**Ship Channel**	**Houston other**	**DFW**	**Ship Channel**	**Houston other**	**DFW**
PM2.5	3	1	3	–	–	–
Ozone	6	10	6	2	2	2
NO_2	4	3	4	2	2	2
SO_2	3	1	0	3	1	0

02.3 How comparisons are tested

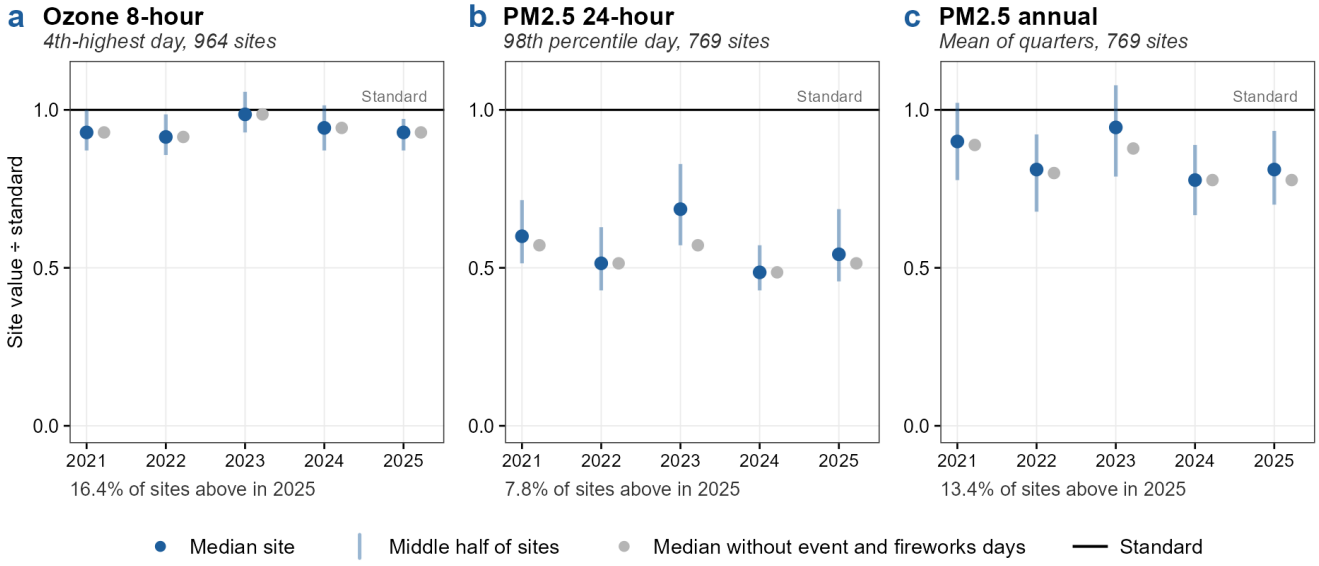
Groups are compared with same-day differences: each day, a group's value is the mean of its reporting sites, and subtracting two groups on the same day cancels most of the weather they share. The median daily difference gets a 95% range from a block bootstrap that resamples 14-day blocks 1,000 times, because neighbouring days are correlated. Each result is then rerun under up to eight checks, including alternative site groups such as the one in TCEQ's 2024 air toxics memo [7], added four-year sites, and dropping event days or the days after a monitor change. A check that leaves the data unchanged is dropped, and a result is said to hold when it keeps its sign with a range that excludes zero. Appendix A.3 shows the code for these steps.

03 National context

03.1 Network and screening

Levels use the sites with data in all five years (964 ozone, 769 PM2.5, 312 NO_2, 301 SO_2 and 135 CO sites), and exceedance counts use all sites. Reporting SO_2 and CO sites fell by 18.1% and 17.4%, and PM2.5 sites grew by 3.9%. The FEM share of PM2.5 rose from 79.2% to 85.2%, with FEM reading 0.3 to 0.5 µg/m^3 above FRM at the same site and day (Figure 2A.1, Appendix).

Each site-year was screened in its standard's own form, rounded as the regulations specify [8–12] (Figure 2A.2). Ozone and annual PM2.5 are missed most often: in 2023, 42.7% of ozone sites and 39.0% of annual PM2.5 sites were above their levels, with the 9.0 µg/m^3 annual PM2.5 level set in 2024 applied to every year. Over three-year windows, which are closer to EPA's design values, 22.8 to 24.1% of ozone sites were above the standard.



Source: EPA AQS daily data, 2021 to 2025, sites with data in all five years. Single-year screening values, not EPA design values.

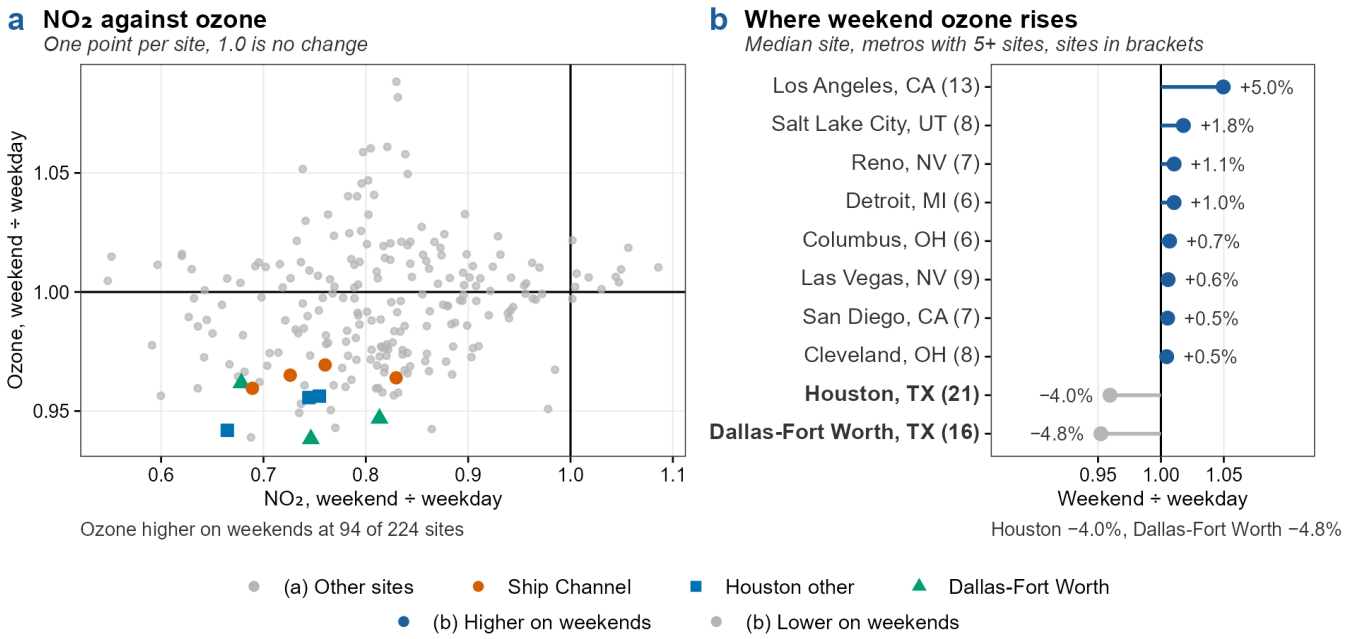
Figure 2A.2. Ozone and annual PM2.5 are missed most often.
Record: national_naaqs_screening.csv

03.2 Events and metro ranks

Events matter much more for PM2.5 than for ozone (Figure 2A.4, Appendix). Labelled smoke, dust and storm days hold 39.9% of PM2.5 exceedance site-days, and 68.2% in 2023, the year of the large Canadian wildfire smoke episodes. Only 14.4% of these event-day exceedances carry EPA's exceptional-event flag. For ozone the share is 1.2%, consistent with ozone exceedances coming mainly from ordinary summer photochemistry. By exceedance site-days, Houston and Dallas–Fort Worth rank 6th and 8th of 353 metros for ozone, against 52nd and 89th of 359 for PM2.5 (Figure 2A.5, Appendix), so ozone is the main problem in both metros.

03.3 Seasons and the weekend effect

Ozone peaks in June at 1.58 times its lowest month and PM2.5 in July at 1.51 times, consistent with summer photochemistry and, for PM2.5, wildfire smoke. NO_2 and CO peak in winter at 1.75 and 1.73 times their lows, consistent with a shallow mixing layer that keeps fresh emissions near the ground. From April to October, weekend NO_2 at the median site falls to 0.805 of its weekday level while ozone stays at 0.987, because ozone does not follow NOx one for one. Where VOCs limit ozone formation, less NO on weekends means less titration, and ozone can rise. Weekend ozone is 5% higher than on weekdays in Los Angeles, consistent with VOC-limited conditions, and 4% and 4.8% lower in Houston and Dallas–Fort Worth, consistent with more NOx-limited summer weekends (Figure 2A.10). Section 5 returns to titration at the hourly scale.



Source: EPA AQS daily data, April to October 2021 to 2025, sites with data in all five years, without event and fireworks days. Weekend: Saturday and Sunday; Memorial Day, Independence Day and Labor Day are left out.

Figure 2A.10. Weekend ozone rises in Los Angeles and falls in Houston and Dallas–Fort Worth.
Record: national_weekend_summary.csv, national_ozone_weekend_metros.csv

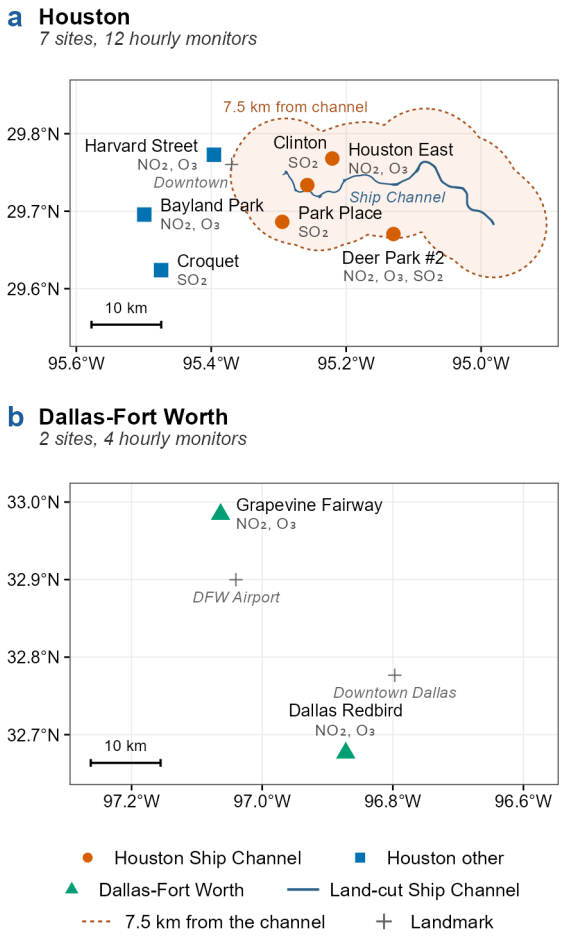
04 Houston case study: daily

This section compares the three site groups using daily values at the sites with data in all five years (Table 2.2). Figure 1C.1 shows the hourly monitors used in Section 5. The daily groups include more Harris County sites, split by the same 7.5 km rule.

04.1 NO_2 and ozone by group

NO_2 is highest at the Ship Channel in all 5 years. The median Ship Channel site's mean daily NO_2 peak is 1.44 ppb above Houston other, or 2.49 ppb when the days after the NO_2 method changes at Park Place (November 2023) and Harvard Street (April 2024) are left out, mostly because Park Place, a Ship Channel site, reads lower on its new method. NO_2 also falls with distance from the channel. With four-year sites added, the Spearman correlation across 10 sites is $\rho = -0.72$ ($p = 0.02$). Dropping one site at a time keeps ρ between -0.83 and -0.62, but without the farthest site (48.9 km away), p rises to 0.077, so the gradient is treated as supporting evidence only.

Ozone shows the opposite pattern. Dallas–Fort Worth has the most ozone exceedance days per site in all 5 years, and the Ship Channel has the fewest in 3 of 5, consistent with fresh NO lowering ozone near the channel (H2).



Source: EPA AQS site coordinates; USACE channel. Shaded: within 7.5 km of the channel.

Figure 1C.1. Nine case-study sites in three groups.
Record: data/reference/case_study_sites.csv

04.2 Same-day differences

Table 4.1 gives the same-day differences described in Section 2.3. The Ship Channel has 1.15 ppb more NO_2 and 0.97 ppb less ozone than Houston other on the same days. More NO_2 and less ozone on the same days is what titration predicts, and Section 5 tests this directly with Ox. Houston as a whole runs 4.00 ppb lower in ozone and 1.77 ppb higher in NO_2 than Dallas–Fort Worth, holding in every run.

PM2.5 and SO_2 show no group gap worth reading. The PM2.5 difference holds in only 3 of 6 runs, consistent with PM2.5 being mostly regional. The SO_2 difference holds in every run, but at 0.13 ppb it sits far below the 2 ppb detection limit, so Section 5 reads SO_2 through spike hours instead. Croquet, in Houston other, has more SO_2 spike days than the average Ship Channel site in all 5 years. Figure 2B.7 in the Appendix shows these results by season, year and check.

Table 4.1. Median same-day differences between groups, with block-bootstrap 95% ranges and the share of check runs in which each result holds its sign with a range clear of zero.

Comparison	Pollutant	Median difference (95% range)	Runs holding
Ship Channel minus Houston other	NO_2	+1.15 (+0.81 to +1.41) ppb	7 of 7
	Ozone	−0.97 (−1.11 to −0.77) ppb	6 of 7
	PM2.5	−0.20 (−0.40 to −0.03) µg/m^3	3 of 6
	SO_2	+0.13 (+0.10 to +0.20) ppb	4 of 4
Houston minus Dallas–Fort Worth	NO_2	+1.77 (+1.26 to +2.26) ppb	7 of 7
	Ozone	−4.00 (−4.87 to −3.27) ppb	6 of 6
	PM2.5	+0.47 (+0.22 to +0.73) µg/m^3	6 of 6

04.3 Events

Most Ship Channel monitors stopped reporting during Hurricane Beryl in July 2024. On average only 19.6% of them reported during the event window, against 58.1% in Houston other and 100% in Dallas–Fort Worth (Figure 2B.6, Appendix). Winter Storm Uri raised PM2.5 to 1.20 to 1.57 times its normal level across the groups.

05 Houston case study: hourly

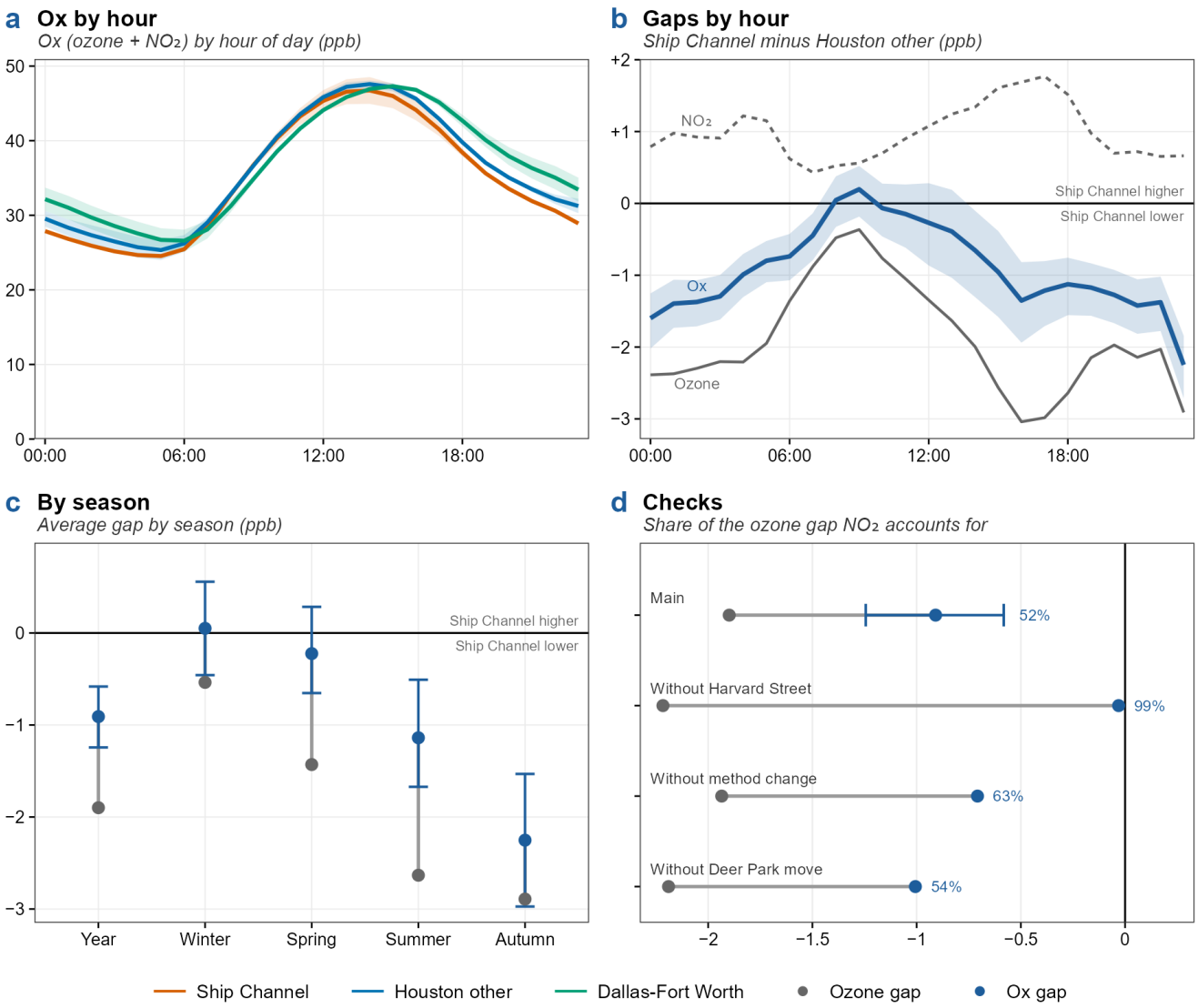
Hourly data show when the group gaps appear and how they depend on wind. Each group has two hourly ozone and NO_2 sites, so these results describe those sites. Harvard Street is the one Houston other site 7.5 to 15 km from the channel, and because it changes the ozone result most, the main numbers are also given without it. All three groups report in the same hour in at least 95% of hours.

05.1 Daily cycles

Ship Channel NO_2 is higher and its ozone lower than Houston other at all 24 hours, under every check. Both gaps are widest late in the day, with NO_2 +1.9 ppb at 17:00 and ozone −2.96 ppb at 16:00 (Figure 2C.2, Appendix). From April to October, 6 to 8 am NO_2 on weekends is 36% below weekdays at the Ship Channel, against 42% in Houston other and 50% in Dallas–Fort Worth. The smaller drop is consistent with more Ship Channel NO_2 coming from sources that run all week, such as industry.

05.2 The Ox balance

Titration ($NO + O_3 \rightarrow NO_2 + O_2$) turns each ozone molecule it destroys into one NO_2 molecule, so it leaves $Ox = O_3 + NO_2$ unchanged. If titration were the only reason Ship Channel ozone is lower, the two groups would have the same Ox. Over the year, Ship Channel ozone is 1.90 ppb lower than Houston other (95% range 1.60 to 2.25), but its Ox is only 0.91 ppb lower, so **Ox closes 52% of the ozone gap** (Figure 2C.4, Table 5.1). Without Harvard Street it closes 99%, it closes at least 52% under every check, and in winter, when little new ozone forms, the main Ox gap's range includes zero. This is consistent with fresh NO from channel sources using up ozone near where it is emitted. With no NO data, Ox cannot be split further into local and regional parts.



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with both pollutants at the same site, 2 sites per group. (b) to (d) Same-hour gaps, 95% ranges from 1,000 resamples of 14-day blocks. Standard-method NO_2 also counts some other nitrogen oxides.

Figure 2C.4. Adding NO_2 back closes at least half the ozone gap.
Record: case_study_ox_checks.csv

Table 5.1. Ship Channel minus Houston other, ppb. Gap closed is the share of the ozone gap that Ox closes.

Hours	Ozone	NO_2	Ox	Gap closed
All hours, all year	−1.90	+0.99	−0.91 (−1.24 to −0.58)	52%
All hours, without Harvard Street	−2.22	+2.19	−0.03	99%
All hours, winter	−0.54	+0.59	+0.05 (−0.46 to +0.56)	109%
Wind from the channel, without Harvard Street	−3.56	+1.41	−2.16 (−2.73 to −1.66)	40%
Other wind, without Harvard Street	−2.35	+2.38	+0.03 (−0.30 to +0.35)	101%

05.3 Calm air and wind

Calm hours, with wind below 0.5 m/s, make up 6.0% of hours at the Ship Channel, 8.9% in Houston other and 1.6% in Dallas–Fort Worth. Compared with windy hours at the same site, hour and month, calm hours carry 3.0 to 6.8 ppb more NO_2 and 3.8 to 9.0 ppb less ozone at every site, each range clear of zero (Figure 2C.8, Appendix). In every group at least 70% of the ozone lost in calm air shows up as NO_2, consistent with titration in still air near the ground.

The Ox gap that titration leaves sits mostly in wind from the channel. Without Harvard Street, Ship Channel Ox is 2.16 ppb lower in wind from the channel but +0.03 ppb in other wind, a gap whose range includes zero (Table 5.1, Figure 2C.11 in the Appendix). This is consistent with ozone forming in channel air as it moves across Houston, so sites downwind of the channel gain Ox. NO_2 also runs 1.4 to 1.6 times its usual level at all four Houston sites in wind from 60 to 90 degrees. A wind direction points toward a source without proving it, since each direction also brings its own kind of weather.

05.4 SO_2 plumes

SO_2 is below the 2 ppb detection limit in 97.2% to 99.4% of hours at three of the four sites, so it is read through spike hours of 5 ppb or more. Spikes are short, with a median length of 1 hour at every site. Croquet, in Houston other, spikes most often, at 5.4 per 1,000 hours. Of its spikes, 96% fall between 9 am and 6 pm, and its summer rate is 4.1 times any other season's, consistent with a raised plume mixed down to the ground by daytime heating. Most of its spike hours, 75%, come from the southwest, a direction that brings only 11% of its hours and points away from the channel (Figure 2C.10, Appendix). At Park Place, 77% of spike hours come from the channel side, which brings 30% of its hours. During Deer Park #2's spike hours, NO_2 is 7.5 ppb above normal (5.1 to 10.0), so the plumes carry nitrogen oxides as well. These patterns support H3: SO_2 arrives in brief, local plumes from more than one direction.

05.5 Robustness

All 10 headline hourly findings hold under every check that applies: without Harvard Street, without its hours after the method change, without Deer Park #2 after its move and without hours on the new wind sensors (Figure 2C.12, Appendix). All 67 bootstrap ranges were redrawn with a second random seed, and none changed whether its range crosses zero.

06 Distributions

Sections 3 to 5 compare average levels. Before these data go into any model, three more questions matter: what shape the daily values take, whether the groups differ in how much ozone swings from day to day, and whether a fitted curve counts high days correctly. At each five-year site, ordinary days are scaled by the site's mean and fitted with three common skewed curves (lognormal, gamma and Weibull), with a normal curve beside them for comparison. All three skewed curves have two parameters, so AIC, a standard score of fit, compares them fairly. Sites where a quarter or more of days sit at or below zero, almost all of them SO_2 sites, are left out, which leaves 2,425 sites. A curve only counts as a clear winner if it keeps winning when the data are resampled in two-week blocks.

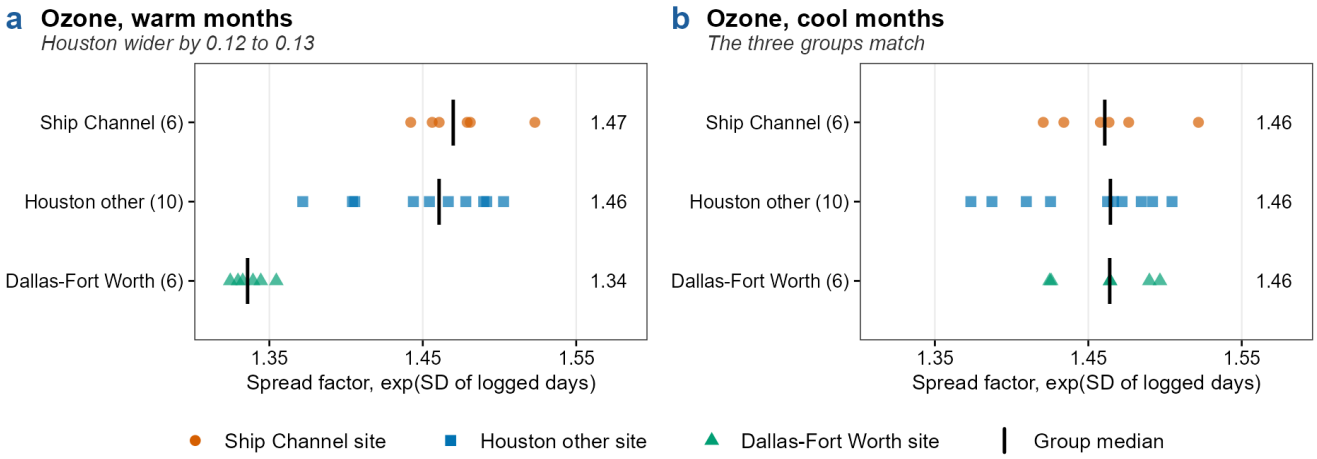
06.1 Shape and fitted curves

Each site has to be read on its own. Pooling all sites together blurs the shape, because much of the spread then comes from differences between sites rather than from one day to the next. At almost every PM2.5 and SO_2 site, and at most CO sites, the values are right-skewed: a few high days sit far above the rest, and a lognormal curve fits better than a normal one. Ozone is close to symmetric, and a normal curve fits better than a lognormal at 82% of ozone sites. NO_2 falls in between, with no single shape winning clearly (Figure 3.5, Appendix).

These curve labels are less certain than they first look. Neighbouring days are linked, so five years of daily data hold less independent information than their size suggests. Once that is allowed for, the best curve for ozone stays the clear winner at only about half of the sites tested. For PM2.5 the monitor matters: the same sites lean much more strongly lognormal on a T640 monitor than on other monitors, consistent with an instrument effect whose cause is not confirmed. In practice, log-scale summaries suit PM2.5, SO_2 and CO, ozone can be summarised on its normal scale, and PM2.5 comparisons across years should note when a site changed monitor.

06.2 Ozone spread in the case study

Figure 3.6 compares how much ozone swings from day to day. The measure is a spread factor, which shows how many times above or below its usual level a site's ozone typically runs, so a larger factor means bigger swings. In warm months, Houston sites swing more than Dallas–Fort Worth sites, by 0.12 to 0.13 on this scale, and both ranges stay clear of zero. The Ship Channel and Houston other match each other, and in cool months no group differs. So the channel lowers ozone's level (Sections 4 and 5) without changing how much it swings. The warm-month gap covers all of Houston, consistent with the city switching between clean Gulf air and still, sunny days, though that cause is not tested here.



Source: EPA AQS daily data, 2021 to 2025, ordinary days, case-study sites, site counts in brackets; cool-season ozone on year-round sites. Warm-month gaps to Dallas-Fort Worth have 95% ranges clear of zero from 1,000 two-week block resamples.

Figure 3.6. Warm-month ozone swings wider at Houston sites.
Record: distribution_fit_checks.csv

06.3 Counting high days

A curve can fit typical days well and still miss the rare high ones, which matters for any model that estimates how often a limit is passed. At most sites, the best-fitting curve misses the number of very high days, those above the site's 99th percentile, by more than 25%. Some of these misses cancel when sites are added together, yet for NO_2 the curves still overcount high days by 54% in total. The choice of curve matters as well. At the ozone standard, the best curves match the measured count, but a lognormal curve would predict about twice as many days above the limit as were actually measured.

For warm-month ozone in the case study, a gamma curve counts days above the standard well in all three groups, within the uncertainty of the measured count. It still overcounts the very highest days, the top 1% at each site, by 51% to 69%. Readings below the detection limit also matter. At Deer Park #2, treating them as "somewhere below the limit" instead of as exact values brings the predicted number of high SO_2 days from about double the measured number down to about right. Fitted curves here describe the data. They should not stand in for counting the actual high days.

06.4 A worked design value

An ozone design value is the three-year average of each year's 4th-highest daily 8-hour value, cut to three decimals. A site meets the standard at 0.070 ppm or below [11]. At Houston Aldine, the 4th-highest days were 0.078, 0.097 and 0.073 ppm in 2023, 2024 and 2025. Their average, 0.0827, is cut to 0.082 ppm, the same value EPA publishes [13]. The site also just meets EPA's rule that at least 90% of days must be valid on average.

Repeating this for all 22 case-study ozone sites over three periods matches EPA at 61 of 66 site-periods. The few misses are each 0.001 ppm low, because the daily cleaning drops short days that EPA keeps when they pass the limit, and none changes whether a site passes. All 22 sites are above the standard for 2023 to 2025, and the highest value is at Deer Park #2, a Ship Channel site. The three groups' typical design values are nearly the same, so the Ship Channel's lower average ozone does not carry over to its highest days. All 12 findings in this section hold under every check that applies (Figure 3.10, Appendix).

07 Limitations

- **No trends.** Five years is too short to separate a trend from year-to-year weather, so changes between years are described but not tested.
- **Small groups.** Each group has two hourly ozone and NO_2 sites, and Houston other has one five-year PM2.5 site, so the hourly results describe those sites rather than whole areas.
- **Pointers, not proof.** Distance to the channel's water stands in for distance to the plants along it, and each wind direction brings its own weather. Both point toward a source without proving it, which is why source claims say "consistent with".
- **No NO or NO_x data.** Ox shows how much of the ozone gap titration explains, but the rest cannot be split into local and regional parts.
- **Near the detection limit.** SO_2 is below 2 ppb in most hours at three of the four sites, and the censored fits use method-wide detection limits, which can differ from a monitor's own.
- **Cleaning choices.** Days with under 75% of hours are dropped, which puts 5 of 66 design values 0.001 ppm below EPA's, and Section 3's single-year screening is not a design value.

Wind is missing for up to 474, 314 and 161 days in a row at Bayland Park, Grapevine Fairway and Deer Park #2.

- **Monitor changes.** PM2.5 moved toward FEM monitors, and Park Place and Harvard Street changed their NO_2 methods in November 2023 and April 2024. Checks drop the affected days, but comparisons between years still partly reflect the monitor mix.
- **Many comparisons.** Dozens of ranges are reported without a multiple-comparison adjustment, so the results that hold under every check are the safest to cite.
- **Curves describe, they do not forecast.** Tail checks test fit on the measured days only.

08 Conclusion and regulatory implications

H1 is supported. Ship Channel NO_2 is higher than Houston other in all 5 years, on the same days (+1.15 ppb, range +0.81 to +1.41) and at all 24 hours, under every check. The distance gradient agrees but rests partly on one far site.

H2 is supported in part. Ship Channel ozone is lower on the same days (−0.97 ppb, range −1.11 to −0.77) and at every hour. Adding NO_2 back closes 52% of the hourly ozone gap, or 99% without Harvard Street, so titration explains at least half of it. The Ox gap that remains sits in wind from the channel, consistent with ozone forming in channel air as it moves downwind.

H3 is supported. SO_2 arrives as spikes with a median length of 1 hour over a background below the detection limit, from the southwest at Croquet (75% of its spike hours) and from the channel side at Park Place (77%), consistent with more than one local source.

These results carry three implications for air quality planning, each limited by the observational design.

Lower ozone near the channel is not cleaner air. Much of the ozone missing near the channel shows up as NO_2, and the Ox left over is consistent with ozone forming downwind. Every case-study ozone site, Ship Channel sites included, is above the 0.070 ppm standard for 2023 to 2025, and the highest design value is at Deer Park #2. This is consistent with treating NO_x near the channel and ozone across Houston as one problem.

SO_2 is best read hourly, at more than one site. One-hour spikes from different directions disappear in daily means and can be missed by a single monitor, which matters for the 1-hour SO_2 standard.

Method choices move the answer. Monitor changes, site groupings and the choice of curve each shift results enough to matter: a default lognormal roughly doubles ozone exceedance counts, and the share of the ozone gap that Ox closes depends on whether Harvard Street is included. Analyses that feed decisions should state these choices and rerun the main result under each one, as this report does.

References

1. U.S. Environmental Protection Agency. NAAQS Table. https://www.epa.gov/criteria-air-pollutants/naaqs-table
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7. Texas Commission on Environmental Quality. Region 12 2024 air monitoring evaluation memo. https://www.tceq.texas.gov/downloads/toxicology/monitoring/air-evaluations/2024/reg12.pdf
8. 40 CFR Part 50, Appendix N (PM2.5). https://www.law.cornell.edu/cfr/text/40/appendix-N_to_part_50
9. 40 CFR Part 50, Appendix S (NO_2). https://www.law.cornell.edu/cfr/text/40/appendix-S_to_part_50
10. 40 CFR Part 50, Appendix T (SO_2). https://www.law.cornell.edu/cfr/text/40/appendix-T_to_part_50
11. 40 CFR Part 50, Appendix U (ozone). https://www.law.cornell.edu/cfr/text/40/appendix-U_to_part_50
12. 40 CFR 50.8 (CO). https://www.law.cornell.edu/cfr/text/40/50.8
13. U.S. Environmental Protection Agency. Air Quality Design Values, Ozone Design Values 2023–2025. https://www.epa.gov/air-trends/air-quality-design-values, file posted 8 June 2026.

Regulation text was read on Cornell's Legal Information Institute; the official version is the eCFR (https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-50).

Appendix

A.1 Data pipeline

Run R/0_setup.R once, then the scripts in the order of Table A.2. Each script reads only the saved outputs of earlier scripts, stops on any failed check, and documents every saved table in data_dictionary.csv. Package versions are pinned in renv.lock. A clean rerun of every script on 13 November 2026 reproduced all 137 outputs byte for byte. Full source code is in the GitHub repository.

Table A.1. Daily cleaning steps, starting from 8,930,992 raw rows.

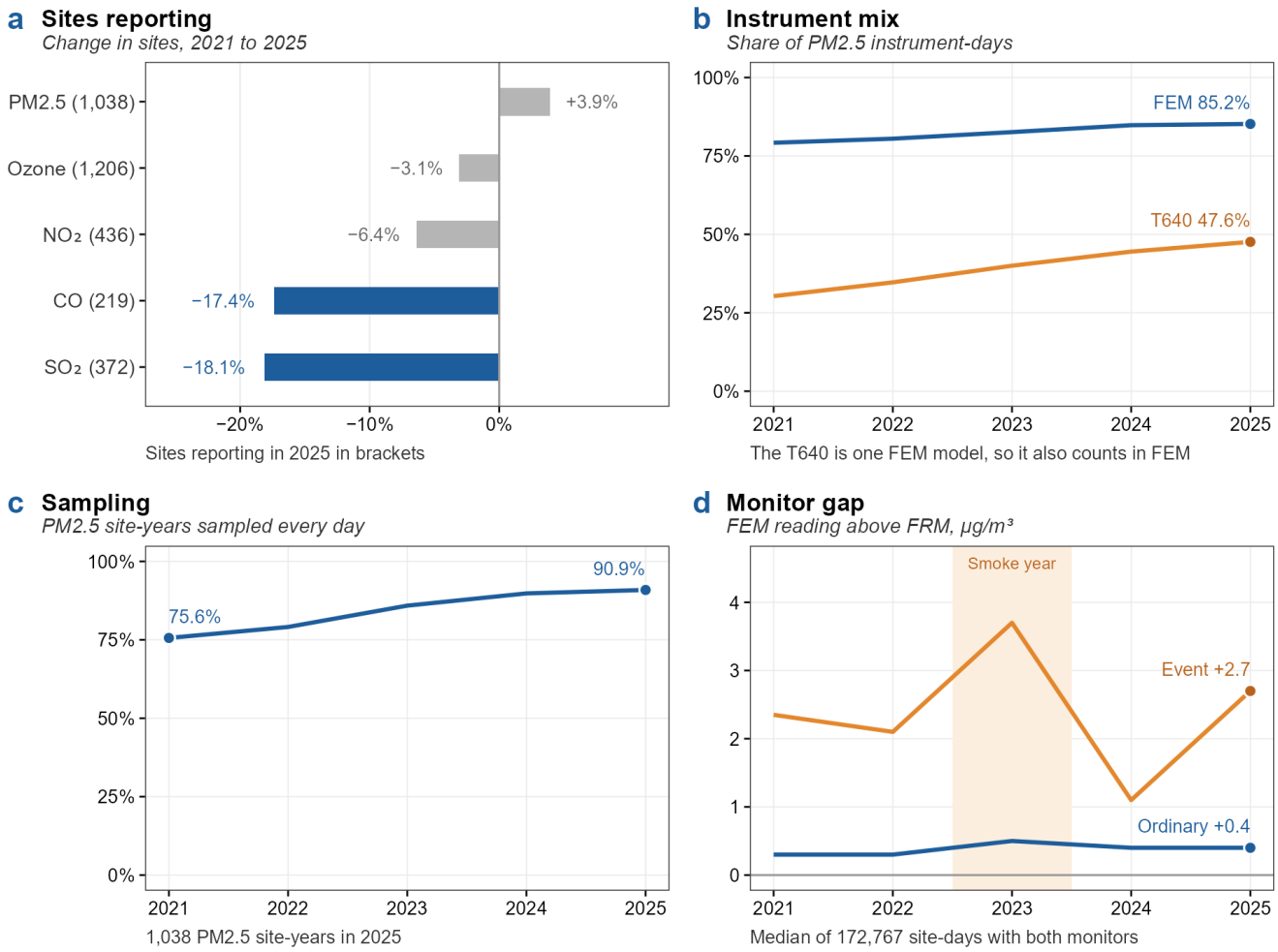
Step	Rows removed	Rows left
Drop non-US sites	9,982	8,921,010
Keep one standard per pollutant	2,956,079	5,964,931
Validate method codes	315,860	5,649,071
Keep event-included values	2,680	5,646,391
Drop T640 corrected rows overlapping firmware data	338	5,646,053
Apply 75% completeness rule	88,478	5,557,575
Drop values outside method range	1	5,557,574

Table A.2. Scripts in run order, with links to the source on GitHub.

Script	What it does	Main outputs	Lines
0_setup.R	Installs the packages and TinyTeX	None	28
utils.R	Shared helpers: chunked reader, NAAQS statistics, saving, chart theme	None	897
1a_import_clean.R	Imports and cleans the 25 daily files	Cleaned daily table, sites, coverage	1,026
1b_event_checks.R	Checks each event label against the data	None	405
1c_site_selection.R	Sets the site groups and case-study monitors	Site list, Figure 1C.1	671
1d_hourly_import_clean.R	Reads the hourly pollutant files in chunks, flags and validates them	Hourly case-study table	1,192
1e_weather_import_clean.R	Reads hourly wind and temperature, sets calm hours and wind sectors	Hourly weather table	1,488
2a_daily_eda.R	National analysis (Section 3)	Figures 2A.1 to 2A.10, national records	2,715
2b_daily_case_study.R	Daily case study (Section 4)	Figures 2B.1 to 2B.8, daily records	2,823
2c_hourly_eda.R	Hourly case study (Section 5)	Figures 2C.1 to 2C.12, hourly records	4,432
3_distributions.R	Distribution shapes, curve fits, tails and design values (Section 6)	Figures 3.1 to 3.10, distribution records	5,289

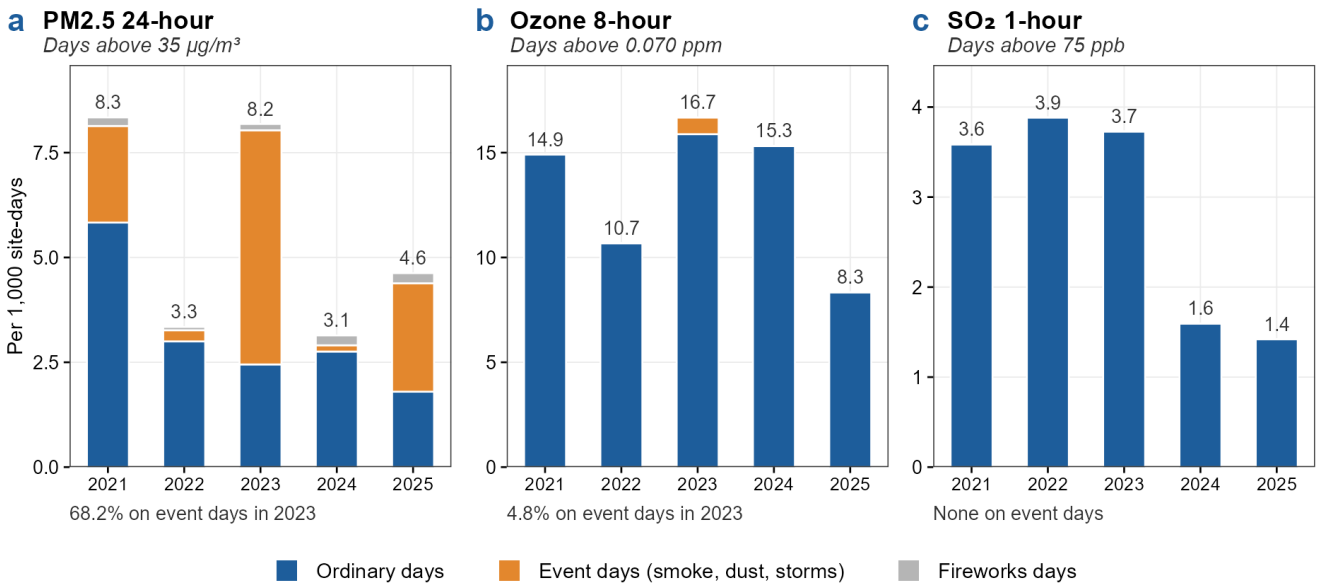
A.2 Additional figures

The 12 figures below support Sections 3 to 6, and each prints the record it is drawn from.



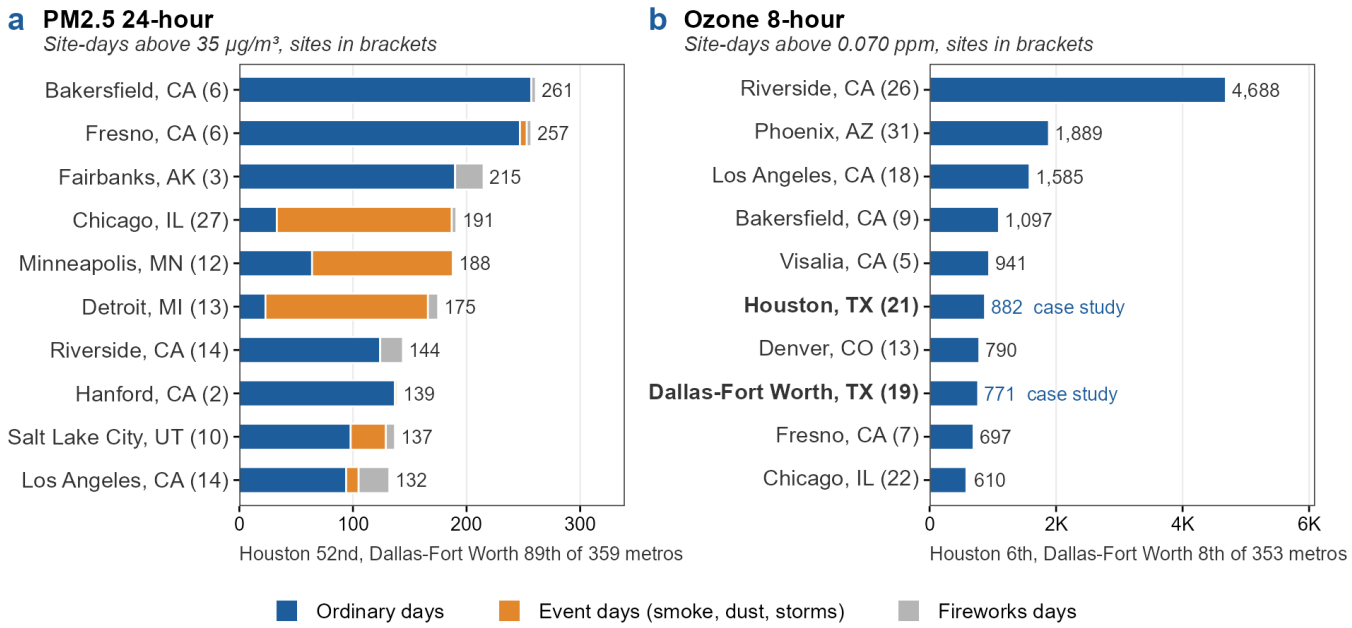
Source: EPA AQS daily data, 2021 to 2025, all sites, after Phase 1 cleaning. Event days are the labelled smoke, dust and storm windows in events.csv; fireworks days are not shown.

Figure 2A.1. SO_2 and CO sites fell as PM2.5 went automated.
Record: national_sites_by_year.csv, national_pm25_methods.csv



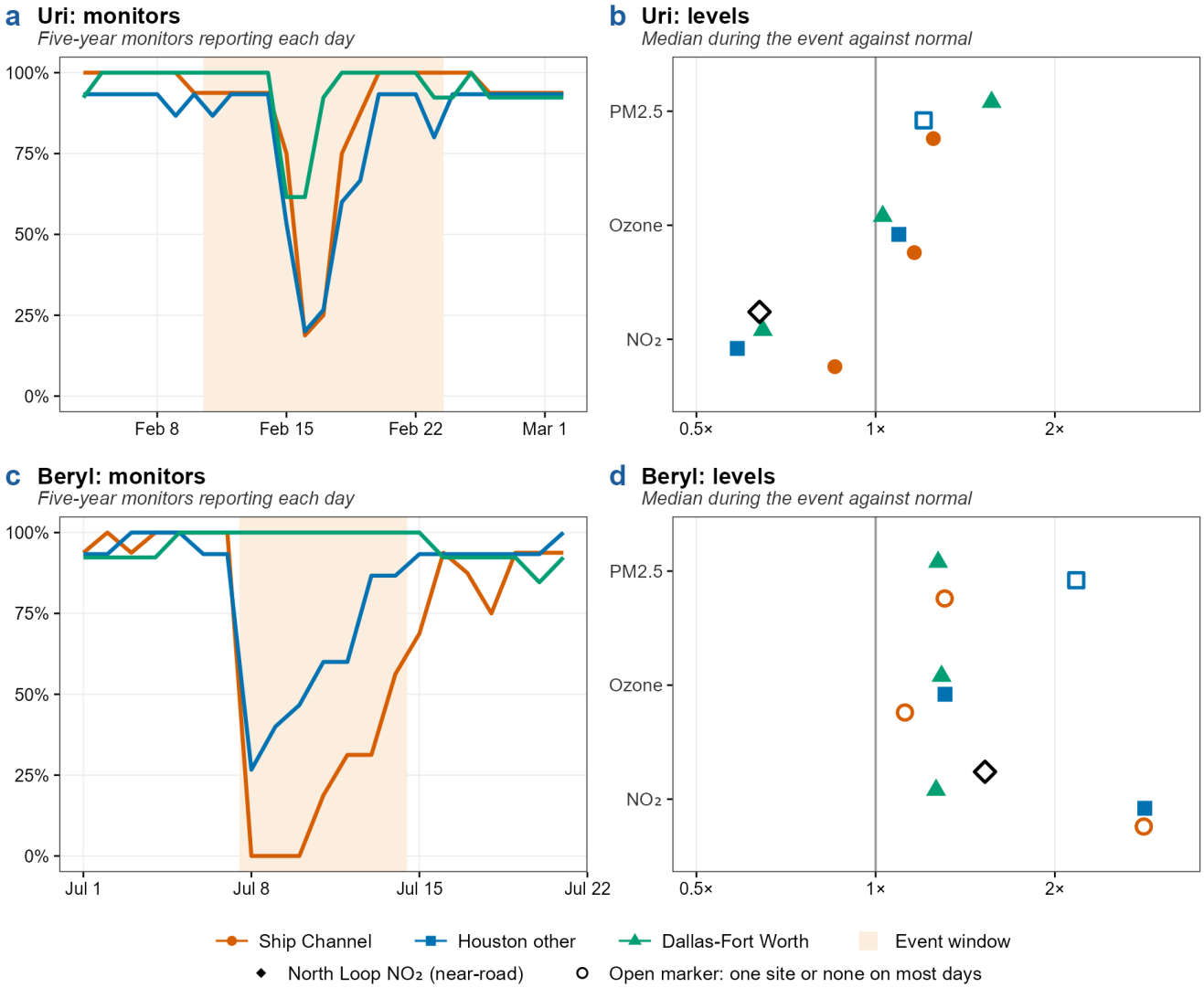
Source: EPA AQS daily data, 2021 to 2025, every site reporting that year. Screening counts, not violations: the standards allow some days above the level. Event days: events.csv.

Figure 2A.4. Event days hold over a third of PM2.5 exceedances.
Record: national_exceedances_by_year.csv



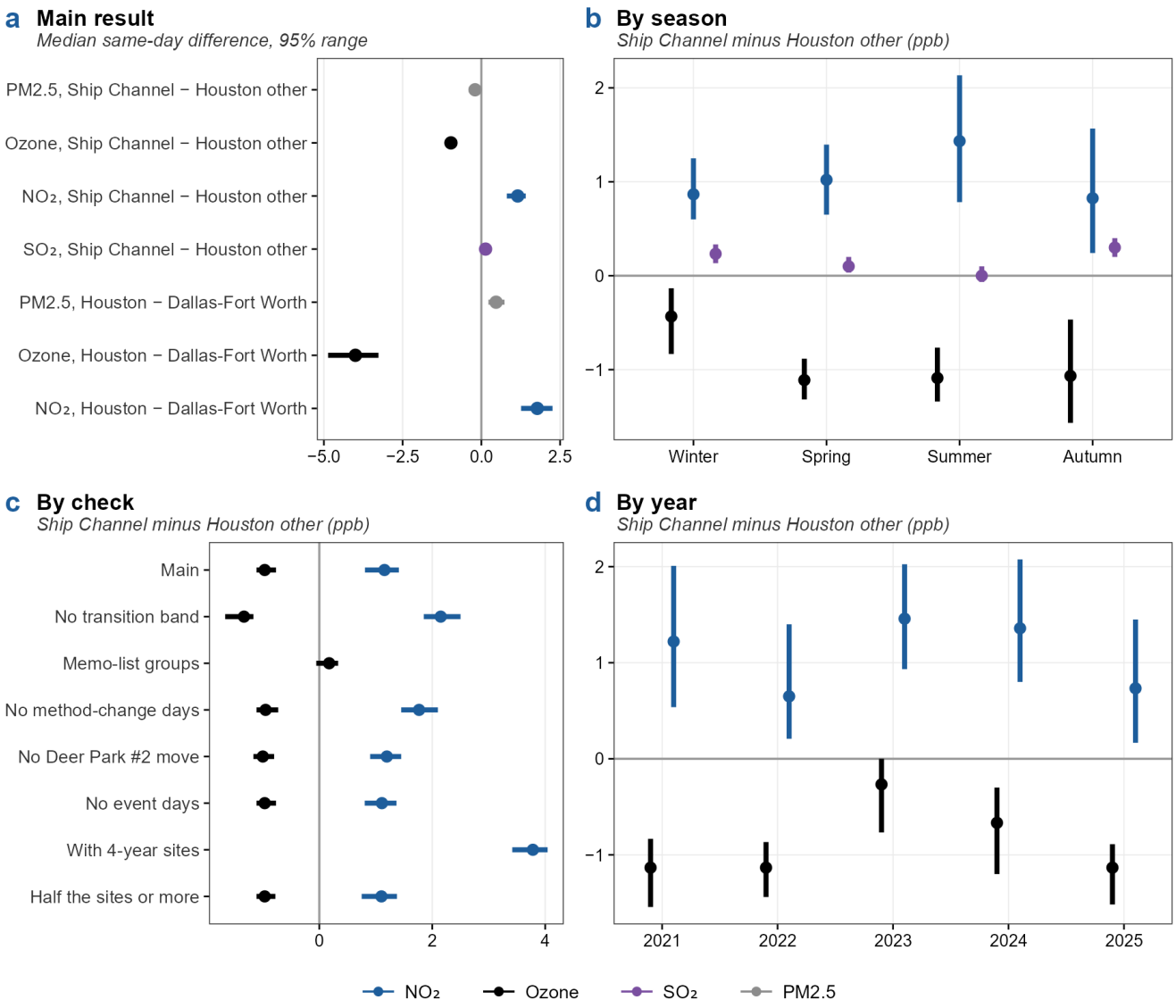
Source: EPA AQS daily data, 2021 to 2025, all sites, by core-based statistical area. Totals grow with the number of sites in a metro. Screening counts, not violations. Bold: the two case-study metros.

Figure 2A.5. California metros top both lists.
Record: national_metro_ranks.csv



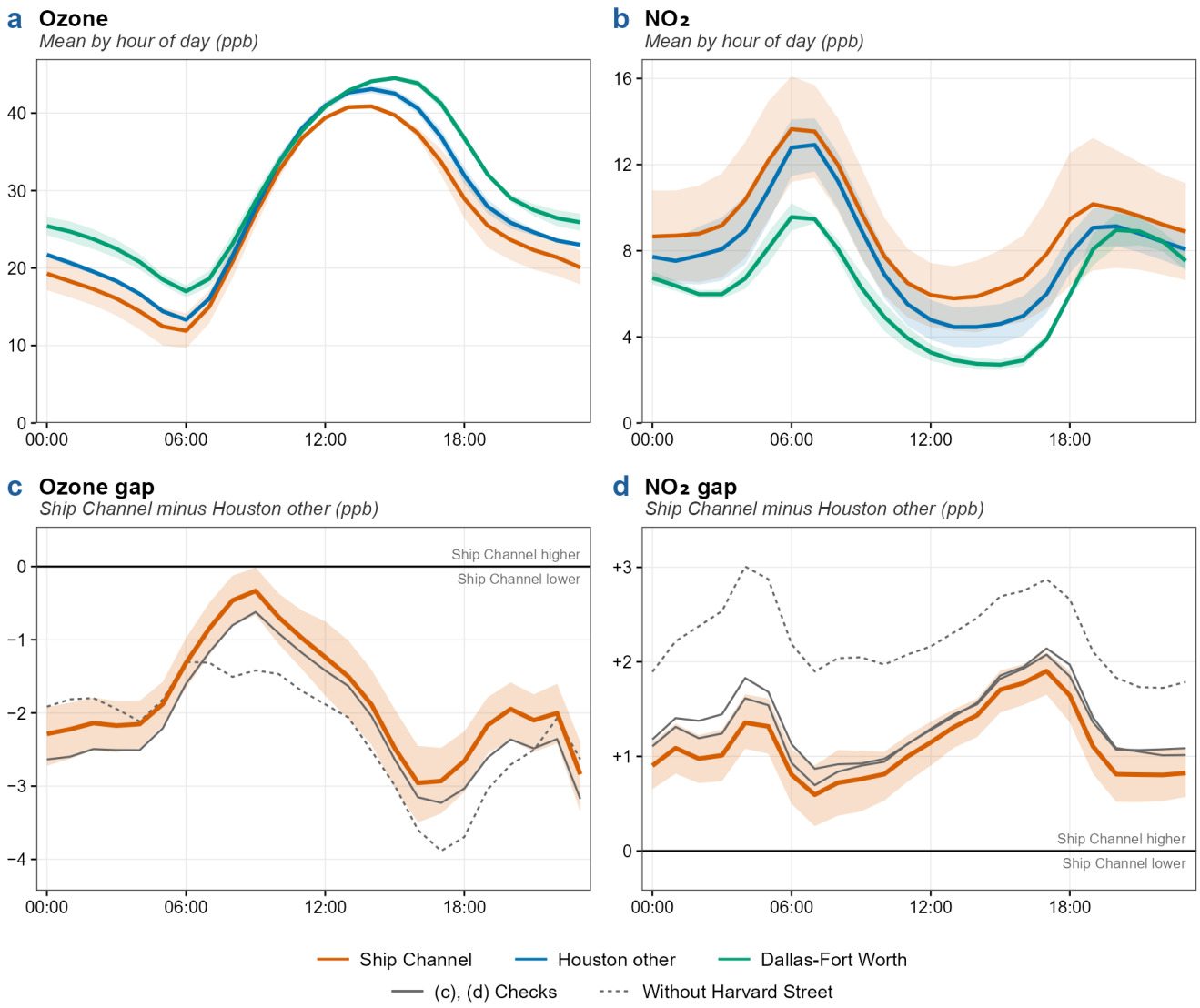
Source: EPA AQS daily data; five-year monitors (site and gas): Ship Channel 16, Houston other 15, Dallas-Fort Worth 13. Shaded: the events.csv window. Normal: the same dates in the other four years. SO_2 sits near zero and is left out of (b) and (d).

Figure 2B.6. Beryl knocked out most Ship Channel monitors.
Record: case_study_event_reporting.csv, case_study_event_levels.csv



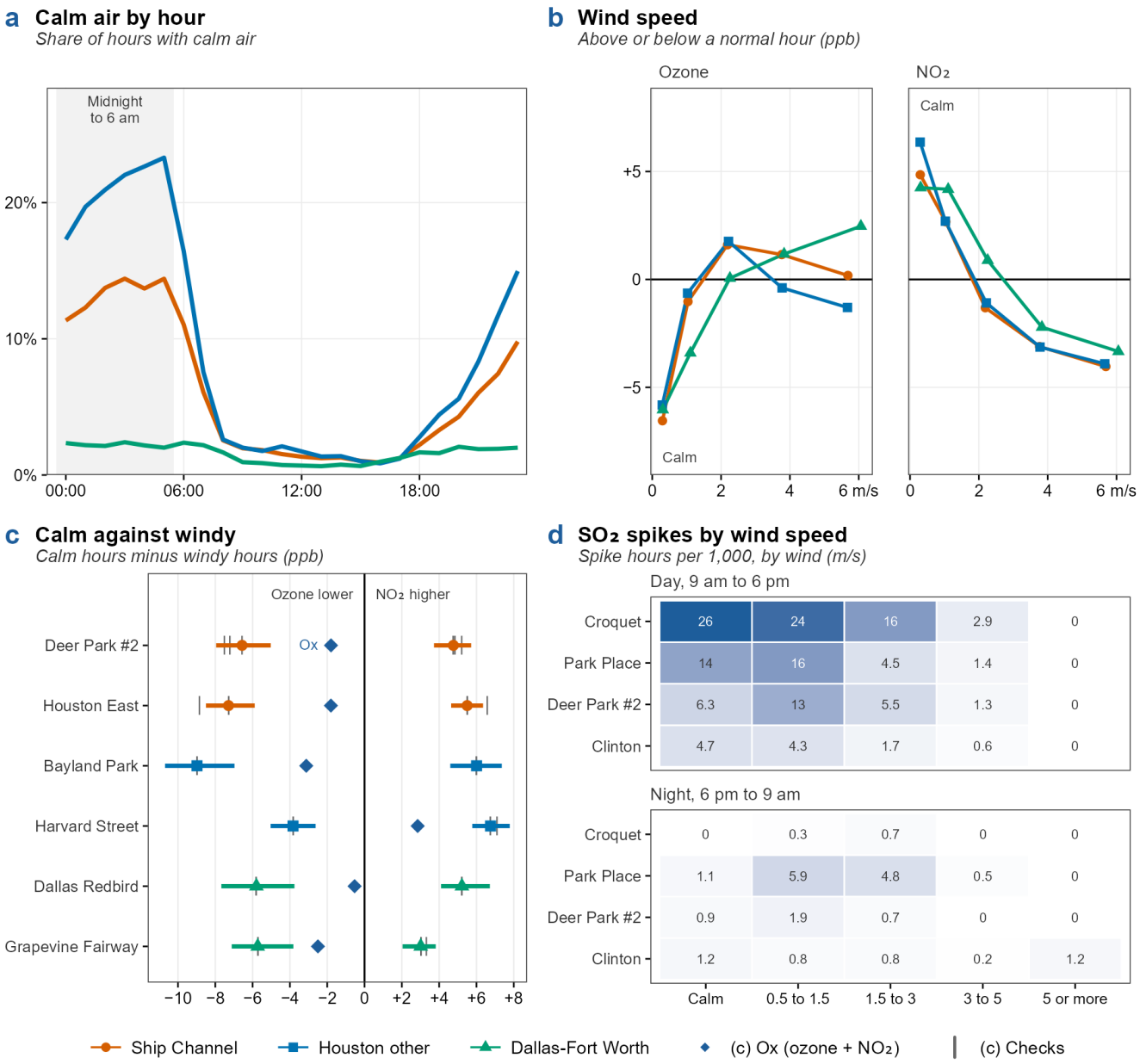
Source: EPA AQS daily data, 2021 to 2025, five-year sites. Group value: the mean of the sites reporting that day; gases in ppb, PM2.5 in µg/m³. 95% ranges: 1,000 resamples of 14-day blocks. (c) Every check from sections 2 to 7.

Figure 2B.7. Same-day NO_2 is higher at the Ship Channel.
Record: case_study_paired_differences.csv



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours, 2 sites per group; local standard time (06:00 is 7 am daylight time). (a), (b) Bands span each group's two sites. (c), (d) 95% ranges from 1,000 resamples of 14-day blocks; every check keeps both gaps at all hours.

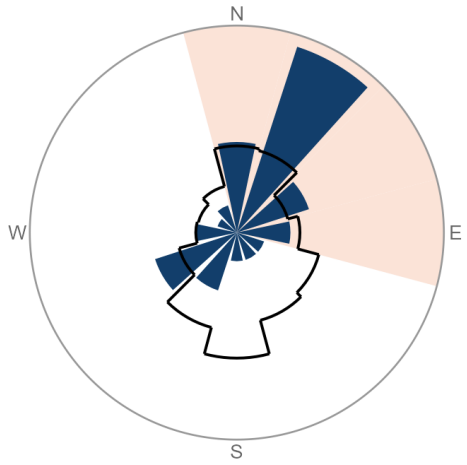
Figure 2C.2. Ship Channel: more NO_2, less ozone, every hour.
Record: case_study_hourly_gaps.csv



Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with wind; calm: under 0.5 m/s. (b), (c) Against hours at the same site, hour and month. (c) 95% ranges from 14-day blocks. (d) Spike: an hour at 5 ppb or more.

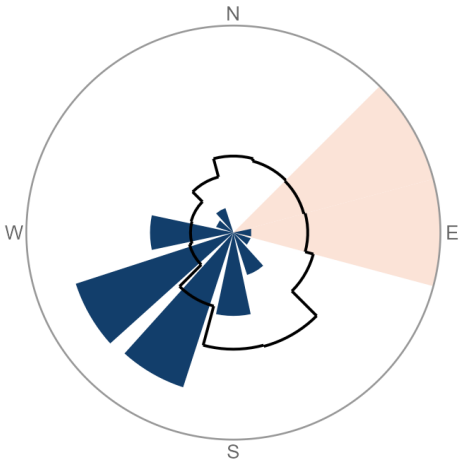
Figure 2C.8. Calm air lifts NO_2 and cuts ozone at every site.
Record: case_study_calm_air.csv

a **Park Place**
Ship Channel



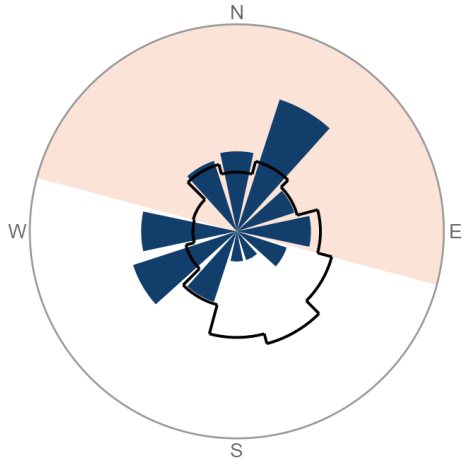
Channel side: 77% of spikes, 30% of hours

b **Croquet**
Houston other



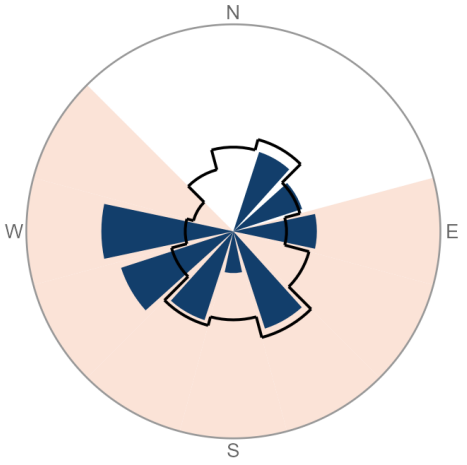
Southwest: 75% of spikes, 11% of hours

c **Deer Park #2**
Ship Channel



Channel side: 56% of spikes, 37% of hours

d **Clinton**
Ship Channel

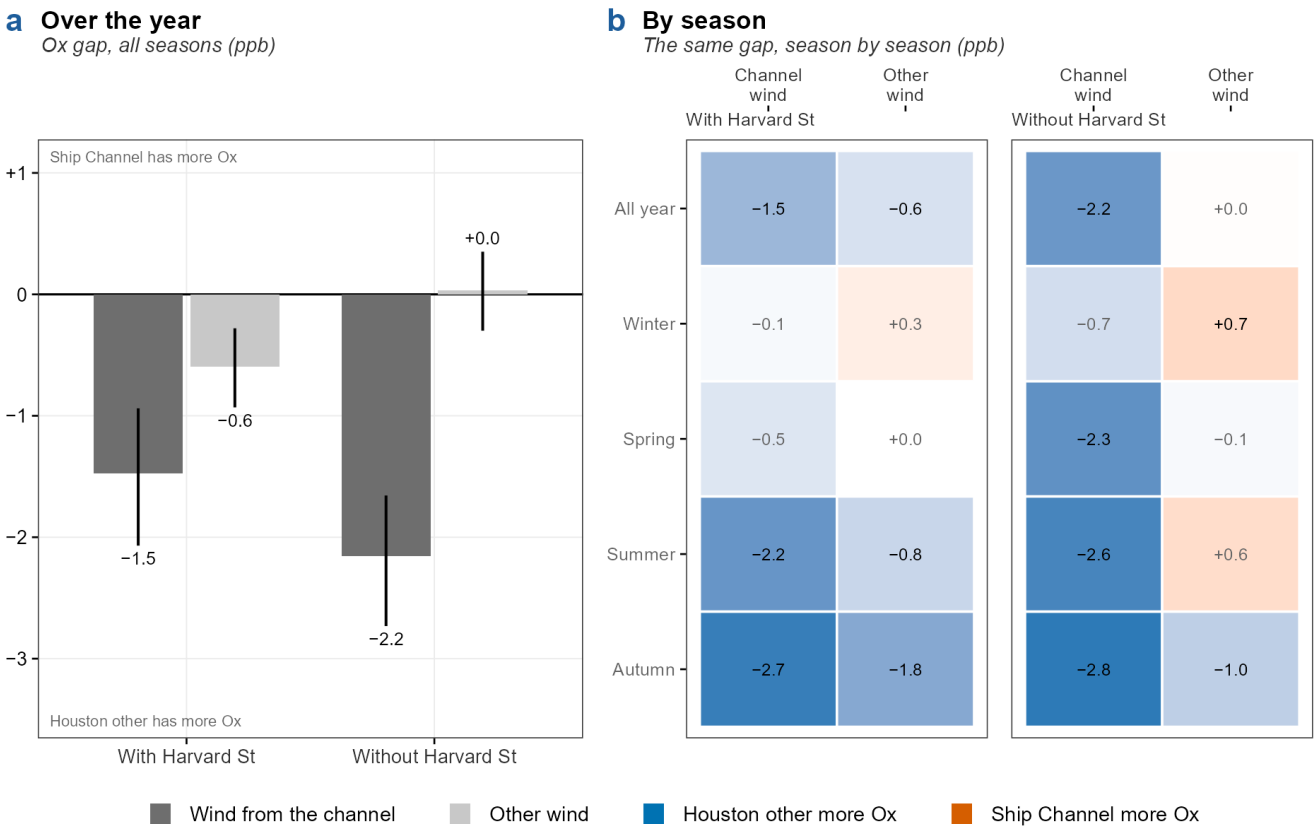


Channel side: 83% of spikes, 65% of hours

Spike hours — All hours — Toward the Ship Channel

Source: EPA AQS hourly data, 2021 to 2025, ordinary hours with a wind direction; 30° sectors. Spike: an hour at 5 ppb or more. Petal area shows each direction's share; past the outline, spikes come from that side more often than the wind does.

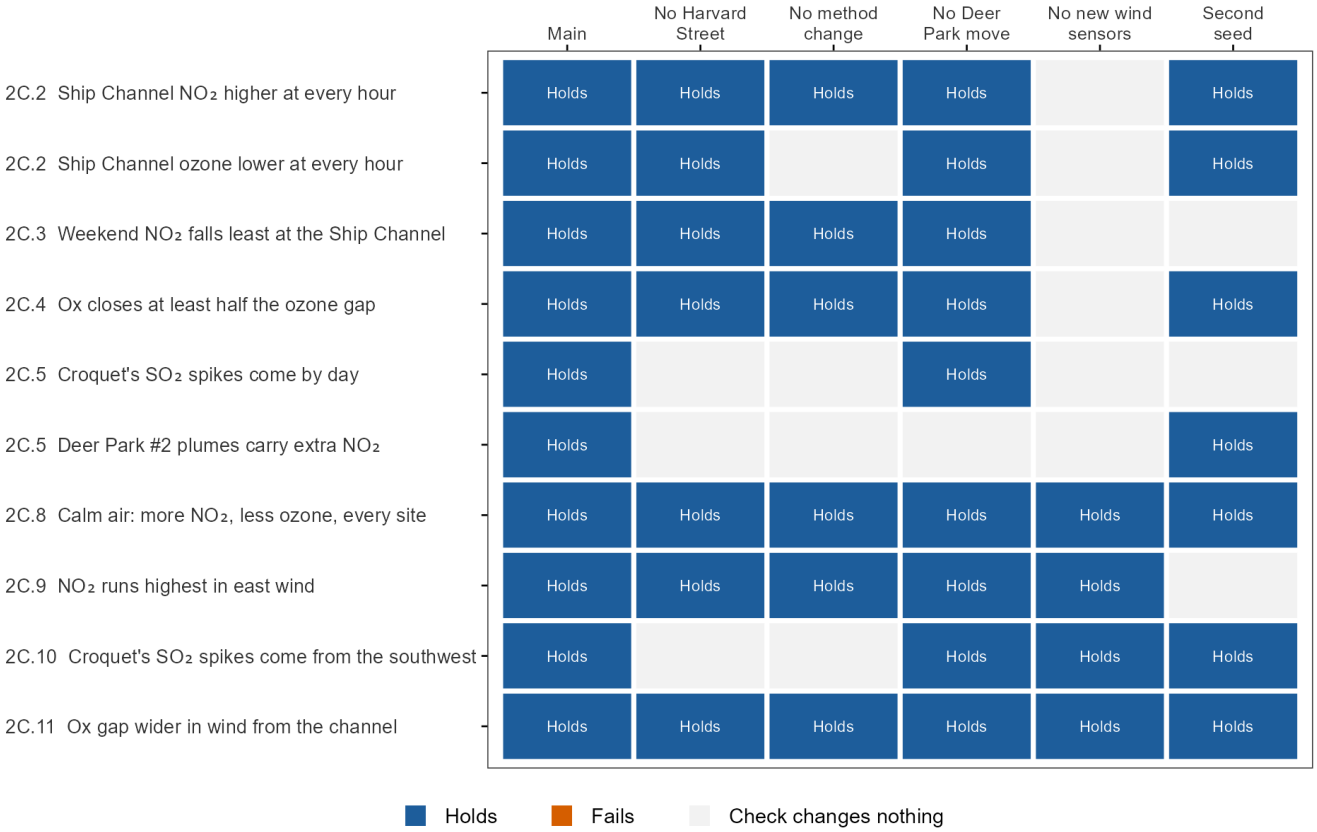
Figure 2C.10. Croquet's SO_2 spikes come from the southwest.
Record: case_study_channel_bearings.csv, case_study_so2_directions.csv



Source: EPA AQS hourly data, 2021 to 2025; wind: mean of the 6 Houston sensors, calm hours left out. Channel wind: 45° to 135°, where the channel lies from Bayland Park and Harvard Street. 95% ranges from 14-day blocks; grey values could be zero.

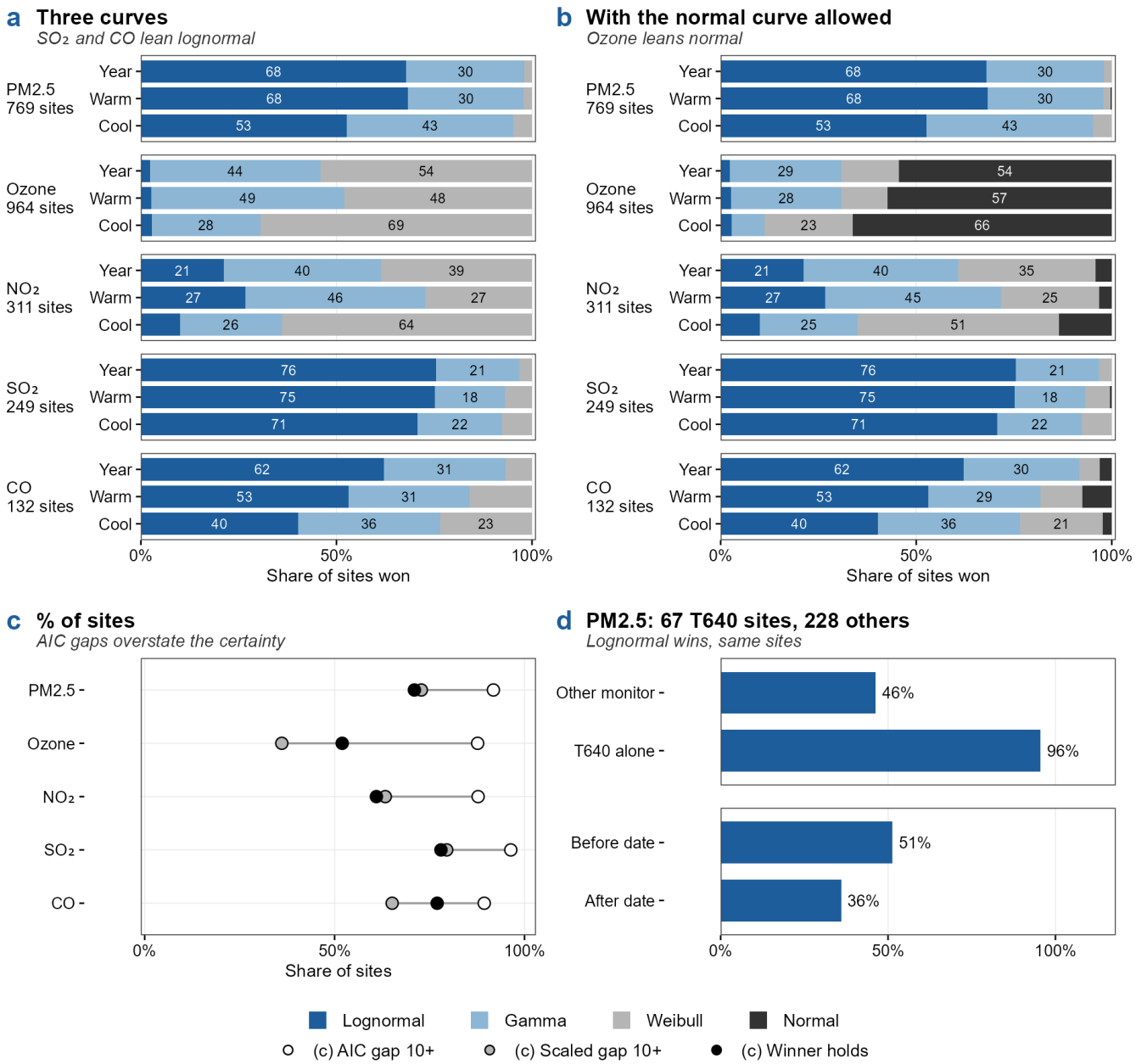
Figure 2C.11. The leftover Ox gap appears in wind from the channel.
Record: case_study_ox_gap_by_wind.csv

Findings and checks
Does each finding survive each check?



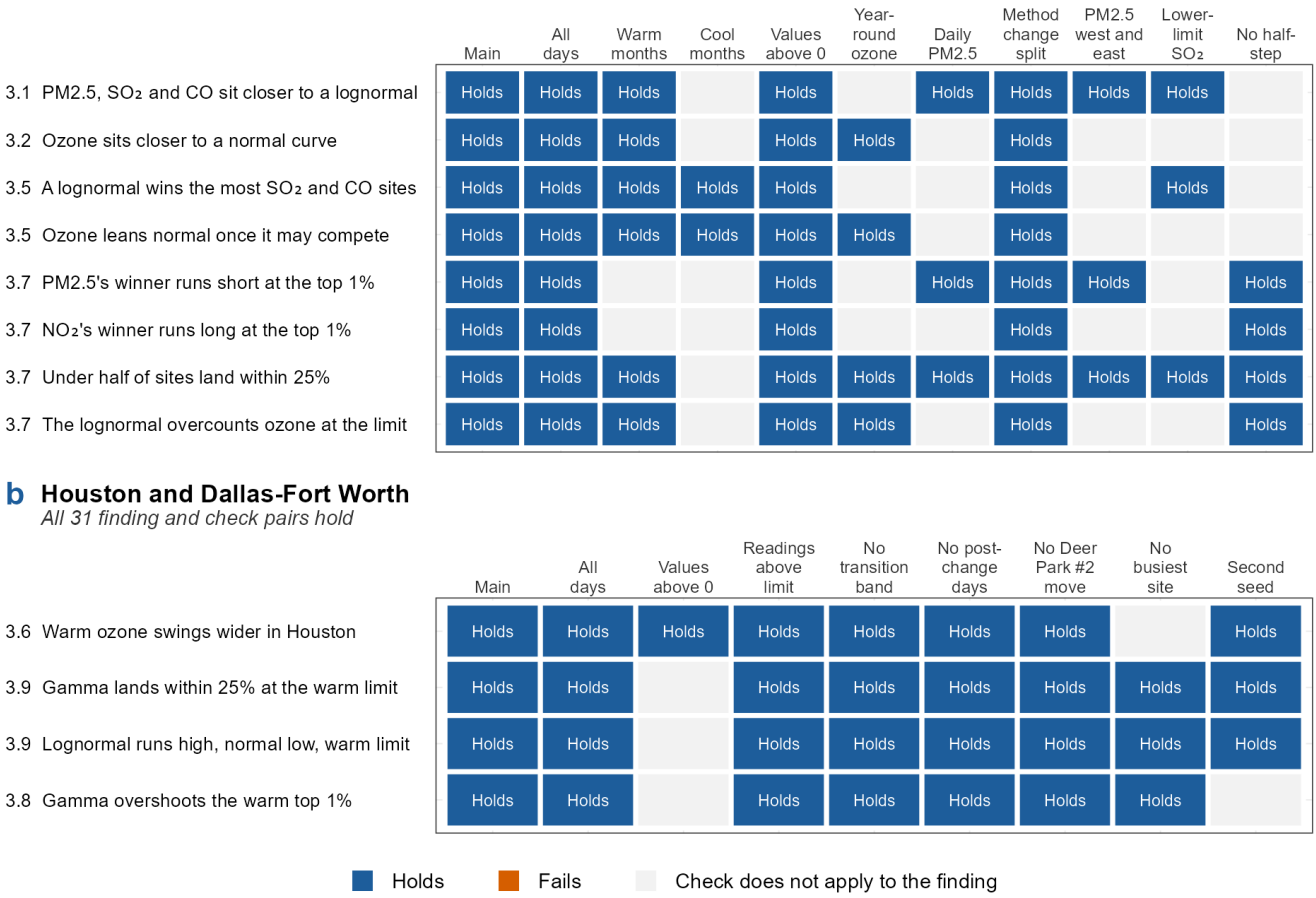
Checks: without Harvard Street, the nearest Houston other site; without its NO_2 after the April 2024 method change; without Deer Park #2 after its May 2025 move; without hours on new wind sensors. Second seed: 67 ranges redrawn with seed 2027.

Figure 2C.12. Every hourly finding holds under every check.
Record: case_study_hourly_robustness.csv, case_study_hourly_seed_check.csv



Source: EPA AQS daily data, 2021 to 2025, ordinary days, five-year sites. Winner: lowest AIC per site. (c) Holds: kept in 90%+ of 100 two-week block resamples of 100 random sites per pollutant. (d) Split at 2023-02-18 for sites without a T640.

Figure 3.5. SO_2 and CO lean lognormal and ozone leans normal.
Record: distribution_fit_summary.csv, distribution_fit_checks.csv



Source: Phase 3 records, EPA AQS daily data 2021 to 2025. (a) five-year sites outside the near-zero rule; (b) warm-month case-study ozone. Second seed: the 95% ranges redrawn with seed 2027.

Figure 3.10. Every distribution finding holds under every check.
Record: distribution_robustness.csv

A.3 Key code

These excerpts are read from the scripts each time the report is built, so they always match the code behind the results.

NAAQS screening (utils.R). Each site-year gets its standard's statistic (a ranked daily value, or the mean of four quarterly means for annual PM2.5), rounded the way the standard specifies, and is compared with the level.

```
naaqs_statistics <- function(site_days, naaqs, naaqs_ranks, percentile_ranks) {
  site_year_ranks <- site_days %>%
    count(pollutant, site_id, year, name = "n_days") %>%
    left_join(naaqs_ranks, by = "pollutant", relationship = "many-to-one") %>%
    left_join(percentile_ranks, by = join_by(percentile, between(n_days, min_days, max_days)),
        relationship = "many-to-one") %>%
    mutate(value_rank = coalesce(fixed_rank, rank)) %>%
    select(pollutant, site_id, year, n_days, value_rank)

  stopifnot(
```

```
    "Every site-year has a rank" = !anyNA(site_year_ranks$value_rank),
    "No rank is beyond the site-year's days" = all(site_year_ranks$value_rank <= site_year_ranks$n_days)
  )

  short_term <- site_days %>%
    arrange(pollutant, site_id, year, desc(value)) %>%
    mutate(value_order = row_number(), .by = c(pollutant, site_id, year)) %>%
    inner_join(site_year_ranks, by = c("pollutant", "site_id", "year"),
        relationship = "many-to-one") %>%
    filter(value_order == value_rank) %>%
    inner_join(filter(naaqs, standard != "Annual") %>% select(pollutant, standard),
        by = "pollutant", relationship = "many-to-one") %>%
    select(pollutant, standard, site_id, year, n_days, value)

  pm25_annual <- site_days %>%
    filter(pollutant == "PM2.5") %>%
    mutate(quarter = quarter(date_local)) %>%
    summarise(quarter_mean = mean(value), .by = c(pollutant, site_id, year, quarter)) %>%
    summarise(n_quarters = n(), value = mean(quarter_mean), .by = c(pollutant, site_id, year)) %>%
    mutate(standard = "Annual") %>%
    left_join(select(site_year_ranks, pollutant, site_id, year, n_days), by = c("pollutant",
        "site_id", "year"), relationship = "one-to-one")

  stopifnot(
    "One short-term value per site-year" = nrow(short_term) == nrow(site_year_ranks),
    "Annual PM2.5 has 1 to 4 quarters per site-year" = all(between(pm25_annual$n_quarters, 1, 4))
  )

  pm25_annual <- pm25_annual %>%
    filter(n_quarters == 4)

  bind_rows(short_term, pm25_annual) %>%
    left_join(select(naaqs, pollutant, standard, level, digits, rounding), by = c("pollutant",
        "standard"), relationship = "many-to-one") %>%
    mutate(value_rounded = if_else(rounding == "truncate", truncate_to(value, digits),
        round_half_up(value, digits))) %>%
    mutate(is_above = value_rounded > level) %>%
    select(pollutant, standard, site_id, year, n_days, n_quarters, value, value_rounded, level,
        is_above)
}
```

Chunked hourly read (utils.R). Each 1-million-row chunk is filtered to the case-study monitors before the next is read, so the full hourly files never sit in memory, and the result is cached until the zip changes.

```
keep_case_study_rows <- function(chunk, monitors) {
  chunk %>%
    filter(`State Code` == "48") %>%
    mutate(site_id = str_c(`State Code`, `County Code`, `Site Num`, sep = "-")) %>%
    semi_join(monitors, by = c("site_id", "Parameter Code" = "parameter_code"))
}

filter_case_study_zip <- function(zip_path, out_dir, monitors) {
  out_path <- file.path(out_dir, str_replace(basename(zip_path), "\\.zip$", ".rds"))

  zip_size <- file.size(zip_path)

  zip_mtime <- as.numeric(file.mtime(zip_path))

  if (file.exists(out_path)) {
    saved <- read_rds(out_path)
    if (identical(saved$zip_size, zip_size) && identical(saved$zip_mtime, zip_mtime)) {
```

```
      return(out_path)
    }
  }

  rows_read <- 0

  filtered_rows <- read_csv_chunked(
    zip_path,
    DataFrameCallback$new(\(chunk, pos) {
      stopifnot("readr reported parsing problems" = nrow(problems(chunk)) == 0)
      rows_read <<- rows_read + nrow(chunk)
      keep_case_study_rows(chunk, monitors)
    }),
    chunk_size = 1e6,
    col_types = aqs_hourly_col_spec
  )

  write_rds(
    list(rows_read = rows_read, data = filtered_rows, zip_size = zip_size, zip_mtime = zip_mtime),
    out_path,
    compress = "gz"
  )

  rm(filtered_rows)
  invisible(gc())

  out_path
}
```

Same-day differences and the block bootstrap (2b_daily_case_study.R). The median daily difference gets a 95% range from resampled 14-day blocks, which keeps the correlation between neighbouring days.

```
block_bootstrap_range <- function(x) {
  n <- length(x)
  if (n < 2 * bootstrap_block_days) {
    return(c(NA_real_, NA_real_))
  }
  block_starts <- seq_len(n - bootstrap_block_days + 1)
  n_blocks <- ceiling(n / bootstrap_block_days)
  medians <- vapply(seq_len(bootstrap_reps), \(rep) {
    picked <- sample(block_starts, n_blocks, replace = TRUE)
    median(x[as.vector(outer(seq_len(bootstrap_block_days) - 1, picked, `+`))][seq_len(n)])
  }, numeric(1))
  unname(quantile(medians, c(0.025, 0.975)))
}

summarise_differences <- function(differences, by) {
  differences %>%
    arrange(date_local) %>%
    summarise(
      n_days = n(),
      median_difference = median(difference),
      pct_days_positive = 100 * mean(difference > 0),
      range = list(block_bootstrap_range(difference)),
      .by = all_of(by)
    ) %>%
    mutate(ci_low = vapply(range, \(r) r[1], numeric(1))) %>%
    mutate(ci_high = vapply(range, \(r) r[2], numeric(1))) %>%
    select(-range)
}
```

Dropping checks that change nothing (2b_daily_case_study.R). A check whose days and values match the main run exactly would only add resampling noise, so it is removed before any range is drawn.

```
difference_series <- daily_differences %>%
  summarise(dates = list(date_local), values = list(difference), .by = c(site_set, contrast,
      pollutant))

main_series <- difference_series %>%
  filter(site_set == "Main") %>%
  select(contrast, pollutant, main_dates = dates, main_values = values)

unchanged_checks <- difference_series %>%
  filter(site_set != "Main") %>%
  inner_join(main_series, by = c("contrast", "pollutant"), relationship = "many-to-one") %>%
  mutate(is_unchanged = mapply(\(d, v, md, mv) identical(d, md) && isTRUE(all.equal(v, mv)), dates,
      values, main_dates, main_values)) %>%
  filter(is_unchanged) %>%
  select(site_set, contrast, pollutant)

print(unchanged_checks, n = Inf)

daily_differences <- daily_differences %>%
  anti_join(unchanged_checks, by = c("site_set", "contrast", "pollutant"))
```

The Ox balance (2c_hourly_eda.R). Ozone, NO_2 and Ox gaps are taken in the same hour, and the share of the ozone gap that Ox closes is one minus the Ox gap over the ozone gap.

```
ox_paired_gaps <- ox_long %>%
  filter(site_group != "Dallas-Fort Worth") %>%
  summarise(group_ppb = mean(value_ppb), .by = c(site_set, measure, site_group, datetime_utc,
      date_local, hour_local, season)) %>%
  mutate(site_group = if_else(site_group == "Houston Ship Channel", "ship_channel_ppb",
      "houston_other_ppb")) %>%
  pivot_wider(names_from = site_group, values_from = group_ppb) %>%
  filter(!is.na(ship_channel_ppb), !is.na(houston_other_ppb)) %>%
  mutate(gap_ppb = ship_channel_ppb - houston_other_ppb)

ox_checks <- ox_season_gaps %>%
  select(site_set, season, measure, n_days, gap_ppb, ci_low, ci_high) %>%
  mutate(measure = str_to_lower(measure)) %>%
  pivot_wider(names_from = measure, values_from = c(gap_ppb, ci_low, ci_high),
      names_glue = "{measure}_{.value}") %>%
  select(site_set, season, n_days, ozone_gap_ppb, no2_gap_ppb, ox_gap_ppb, ozone_ci_low,
      ozone_ci_high, ox_ci_low, ox_ci_high) %>%
  mutate(share_of_ozone_gap_closed = 1 - ox_gap_ppb / ozone_gap_ppb) %>%
  mutate(across(where(is.double), \(x) round(x, 3))) %>%
  arrange(season, site_set)
```